Supplementary: Notation Reference

This supplement provides a comprehensive reference for the mathematical notation used throughout the Cognitive Integrity Framework (CIF) manuscript, including the eusocial and colony cognitive security extensions. Symbols are organized by domain, with cross-references to their formal definitions in the main text and supplements.

Adversary Model Notation I

Adversary Model Notation I

Symbol	Meaning	Defined In
Ω_k	Adversary class k (e.g., Ω_1 = External, Ω_5 = Systemic)	`def:adversary-class`
$\mathcal{R}$	Attack resource tuple $\langle R_C, R_K, R_A, R_P, R_{Co}\rangle$	`def:resources`
R_C	Computational resources (FLOPS-hours)	`tab:resource-types`
R_K	Knowledge resources (system understanding)	`tab:resource-types`
R_A	Access resources (available channels)	`tab:resource-types`
R_P	Persistence resources (temporal presence)	`tab:resource-types`
R_{Co}	Coordination resources (multi-party synchronization)	`tab:resource-types`
D_{score}	Detectability score of an attack	`def:detectability`
$\mathcal{C}_{\text{adv}}$	Adversarial capability set	`def:capability-set`
$\mathcal{K}_{\bullet}$	Adversary knowledge component of the class tuple; the subscript names its scope ($\mathcal{K}_{\text{public}}$, $\mathcal{K}_{\text{domain}}$, $\mathcal{K}_{\text{protocol}}$, $\mathcal{K}_{\text{internal}}$)	`sec:threat-model`
$\mathcal{E}$	Communication edge set $\{(a_i, a_j) : \mathcal{C}(a_i, a_j) = 1\}$; the adversary-controlled subset is $\mathcal{E}_{\text{ctrl}} \subseteq \mathcal{E}$ (distinct from the stigmergic $\mathcal{E}$ below)	`def:omega4`
$\mathcal{A}_{\text{BIM}}$	Belief injection/manipulation attack class	`sec:attack-taxonomy`
$\mathcal{A}_{\text{BI}}$	Belief injection attack	`thm:belief-injection`

System Model Notation I

System Model Notation I

Symbol	Meaning	Defined In
$\mathcal{O}$	Multiagent operator tuple $\langle \mathcal{A}, \mathcal{C}, \mathcal{S}, \mathcal{P}, \Gamma \rangle$	`def:multiagent-operator`
$\mathcal{A}$	Set of agents $\{a_1, \ldots, a_n\}$	`tab:operator-components`
a_i	Individual agent i	`def:multiagent-operator`
n	Number of agents	Throughout
$\mathcal{C}$	Communication adjacency matrix	`tab:operator-components`
$\mathcal{S}$	Shared global state	`def:multiagent-operator`
$\mathcal{P}$	Permission mapping	`def:permission-layer`
Γ	Coordination protocol	`def:multiagent-operator`
σ_i	Cognitive state of agent a_i: $\langle \mathcal{B}_i, \mathcal{G}_i, \mathcal{I}_i, \mathcal{H}_i \rangle$	`def:cognitive-state`
$\mathcal{B}_i$	Belief distribution of agent a_i: $\Phi \to [0, 1]$	`tab:cognitive-components`
$\mathcal{G}_i$	Goal set of agent a_i	`tab:cognitive-components`
$\mathcal{I}_i$	Intention set (committed actions)	`tab:cognitive-components`
$\mathcal{H}_i$	Interaction history	`tab:cognitive-components`
S^t	Global system state at time t	`def:system-state`
σ_i^t	Cognitive state of agent i at time t	`def:cognitive-state`
Φ	Set of propositions	`sec:notation`
ϕ, ψ	Individual propositions	`sec:notation`
$\mathcal{M}$	Message space (distinct from the set of marker types $\mathcal{M}$ of `def:stigmergic-operator`)	`def:firewall`
m	Individual message	`def:firewall`
Σ	OODA input alphabet $\mathcal{M} \cup$ Events (distinct from the stigmergic update function Σ of `def:stigmergic-operator`)	`def:ooda-state`

Trust Calculus Notation I

Trust Calculus Notation I

Symbol	Meaning	Defined In
$\mathcal{T}_{i \to j}$	Trust score from agent i to agent j	`def:trust-function`
$\mathcal{T}_{\text{base}}$ / T_{base}	Base architectural trust (role-based)	`tab:trust-components`
$\mathcal{T}_{\text{rep}}$ / T_{rep}	Reputation trust (historical accuracy)	`tab:trust-components`
$\mathcal{T}_{\text{ctx}}$ / T_{ctx}	Contextual trust (task-specific)	`tab:trust-components`
α, β, γ	Trust component weights ($\alpha + \beta + \gamma = 1$)	`eq:trust-function`
δ	Trust decay factor ($\delta \in (0, 1)$)	`def:trust-delegation`
d	Delegation depth	`def:trust-delegation`
$\mathcal{T}^{\text{del}}$	Delegated trust value	`def:trust-delegation`
$\mathcal{T}^{\text{path}}$	Path trust value	`def:path-trust`
η_m	Modality reliability factor	`def:modality-trust`
η	Learning rate (reputation update)	Trust configuration
ρ	Penalty factor (failure penalty multiplier)	Trust configuration
$\otimes$	Trust delegation operator	`def:trust-algebra`
$\oplus$	Trust aggregation operator	`def:trust-algebra`

Defense Mechanism Notation I

Defense Mechanism Notation I

Symbol	Meaning	Defined In
$\mathcal{F}(m)$	Cognitive firewall classification function	`def:firewall`
D_{inj}	Injection detection score	`def:firewall-rules`
D_{sus}	Suspicious content score	`def:firewall-rules`
τ_1	Firewall reject threshold	`eq:firewall-rules`
τ_2	Firewall quarantine threshold	`eq:firewall-rules`
$\mathcal{B}_{\text{verified}}$	Set of verified beliefs	`def:sandbox`
$\mathcal{B}_{\text{provisional}}$	Set of provisional (sandboxed) beliefs	`def:sandbox`
$\pi(\phi)$	Provenance chain for belief ϕ	`def:evidence`
$V(\pi)$	Provenance verification function	`sec:integrity-properties`
$\mathcal{W}$	Set of canary beliefs (tripwires)	`def:canary`
ω_j	Individual canary belief	`eq:canary-set`
p_j^{exp}	Expected probability for canary j	`eq:canary-set`
ϵ_{drift}	Drift detection threshold	`eq:tripwire-alert`
$\mathcal{I}_{\text{inv}}$	Set of behavioral invariants	`def:invariant-set`
I_k	Individual invariant predicate	`eq:invariant-set`
κ	Corroboration threshold	`sec:belief-update-rules`
TTL	Time-to-live for provisional beliefs	Sandbox configuration

Detection & Analysis Notation I

Detection & Analysis Notation I

Symbol	Meaning	Defined In
S_{drift}	Drift score (belief change magnitude)	`def:drift-score`
D_{KL}	Kullback-Leibler divergence (drift detection)	`def:drift-detection`
w	Sliding window size	`def:drift-detection`
g_t	CUSUM cumulative statistic at time t	`def:cusum-detector`
ν	CUSUM allowance (reference value)	`def:cusum-detector`
h	CUSUM decision interval (alert threshold on g_t)	`def:cusum-detector`
λ	Max delta weight in drift scoring	`eq:drift-score`
λ	Chernoff tilting parameter, $\lambda \in [0, 1]$ (unrelated to the drift weight above and to the colonial-trust decay constant)	`eq:chernoff`
S_{dev}	Behavioral deviation score	`def:deviation-score`
f_k	Feature extractor function	`eq:deviation-score`
μ_k, σ_k	Feature mean and standard deviation	`eq:deviation-score`
AUC	Area Under the ROC Curve	`def:auc`
$\text{TPR}(\tau)$	True Positive Rate at threshold τ	`eq:tpr`
$\text{FPR}(\tau)$	False Positive Rate at threshold τ	`eq:fpr`
$\text{FNR}(\tau)$	False Negative Rate at threshold τ	`eq:threshold-opt`
S_{fused}	Fused detector score	`def:score-fusion`
D_{fused}	Fused detector decision	`def:decision-fusion`
$\text{taint}(\phi)$	Provenance tags for belief ϕ	`def:taint-label`
$\Lambda(m)$	Likelihood ratio $P_{\text{attack}}(m)/P_{\text{benign}}(m)$ (capital Λ; unrelated to λ)	`def:hyp-test`

Consensus & Coordination Notation I

Consensus & Coordination Notation I

Symbol	Meaning	Defined In
q	Quorum threshold for consensus	`def:quorum`
f	Maximum number of Byzantine/compromised agents	`thm:byzantine-req`
$\mathcal{B}_{\text{consensus}}$	Consensus belief function	`def:cog-byzantine`
$\mathcal{D}$	Set of defense mechanisms	`def:defense-composition`
$\mathcal{D}_1 \circ \mathcal{D}_2$	Series defense composition	`eq:series-comp`
$\mathcal{D}_1 \parallel \mathcal{D}_2$	Parallel defense composition	`eq:parallel-comp`
P_{detect}	Detection probability	`eq:series-detection`
r_f	Firewall detection rate	`thm:belief-injection`
r_s	Sandbox verification rate	`thm:belief-injection`

Cost & Performance Notation I

Cost & Performance Notation I

Symbol	Meaning	Defined In
C_{total}	Total defense cost	`def:defense-cost`
C_{compute}	Computational cost	`tab:cost-components`
C_{latency}	Latency cost	`tab:cost-components`
C_{fp}	False positive cost	`tab:cost-components`
$C_{\mathsf{FP}}, C_{\mathsf{FN}}$	Cost of false positive / false negative	`def:cost-threshold`
B_{total}	Total defense benefit	`def:defense-benefit`
L_{CIF}	CIF latency overhead	`thm:latency-overhead`
L_d	Latency of defense d	`eq:latency-budget`
$L_{\max}$	Maximum allowed latency	`eq:latency-budget`

Information & Complexity Notation I

Information & Complexity Notation I

Symbol	Meaning	Defined In
$H(\mathcal{A})$	Entropy of attack $\mathcal{A}$	`thm:min-entropy`
$I(D;\mathcal{A})$	Mutual information between detector and attack	`def:detector-gain`
C_{channel}	Channel capacity	`thm:stealth-impact`
$O(\cdot)$	Big-O complexity bound	`sec:complexity-bounds`
S_{total}	Total space complexity	`eq:total-space`
T_{msg}	Per-message processing time	`eq:message-processing`

Stigmergic & Colony Notation (Supplementary) I

Stigmergic & Colony Notation (Supplementary) I

Symbol	Meaning	Defined In
$\mathcal{O}_\Sigma$	Stigmergic operator tuple	`def:stigmergic-operator`
$\mathcal{E}$	Environmental state (markers/signals)	`def:stigmergic-operator`
Σ	Stigmergic update function (distinct from the OODA input alphabet Σ of `def:ooda-state`)	`def:stigmergic-operator`
$\mathcal{L}$	Set of locations	`def:stigmergic-operator`
$\mathcal{M}$	Set of marker types (distinct from the message space $\mathcal{M}$ of `def:firewall`)	`def:stigmergic-operator`
$\mathcal{N}$	Cyberphysical niche	`def:cyberphysical-niche`
$\mathcal{F}_c$	Emergent collective function	`def:emergent-function`
$\mathcal{T}_c$	Colonial trust function (environment-mediated)	`def:colonial-trust`
$\rho(m, l, t)$	Signal reliability at location l for marker m at time t	`eq:colonial-trust`
λ	Temporal decay constant (colonial trust)	`eq:colonial-trust`
Q_α	Cognitive quorum function for action α	`def:cognitive-quorum`
$\mathcal{A}_e$	Emergent attack	`def:emergent-attack`
CCS	Colony CogSec Score	`def:cogsec-score`
DR_c	Colony-level detection rate	`eq:ccs`
FPR_c	Colony-level false positive rate	`eq:ccs`

General Mathematical Notation I

General Mathematical Notation I

Symbol	Meaning	Usage
$P(\cdot)$	Probability measure	Throughout
$\mathbb{1}[\cdot]$	Indicator function	`eq:decision-fusion`
τ	Generic threshold parameter	Throughout
θ_{drift}	KL drift-detection threshold; distinct from the per-canary deviation tolerance ϵ_{drift}	`eq:drift-detection`
θ_{total}, θ_{step}	Cumulative and per-step drift thresholds in a progressive-drift attack	`def:progressive-drift`
θ (alarm)	Baseline drift alarm threshold $\mu_{\text{baseline}} + k\,\sigma_{\text{baseline}}$	`prop:drift-threshold`
θ_i	Escalation threshold at stage i of the multi-stage detection pipeline	`def:multi-stage`
θ (fusion)	Learned parameters of the fusion network — not a threshold	`eq:learned-fusion`
r_θ	Fisher-Rao displacement corresponding to θ_{drift}	`rem:geometric-bound`
ϵ	Small constant (error rate, deviation)	Throughout
t	Time index	Throughout
$\models$	Satisfaction relation (state satisfies predicate)	`eq:invariant-check`
$\vdash$	Logical entailment	`eq:consistency-def`
$\perp$	Logical contradiction	`eq:consistency-def`
$\perp$	Undecided / undefined	`eq:cog-byzantine`
$\checkmark$	Verification passed	`tab:mc-results`

CTL Temporal Logic Notation (Formal Verification) I

CTL Temporal Logic Notation (Formal Verification) I

Symbol	Meaning	Defined In
AG	"Always globally" (CTL operator)	`eq:ctl-safety`
AF	"Always eventually" (CTL operator)	`eq:ctl-liveness`
EX	"Exists next" (CTL operator)	`sec:temporal-properties`
$\Rightarrow$	Logical implication	Throughout
$\leftrightarrow$	Logical biconditional	Throughout